



A data-driven framework for incentivising fuel-efficient driving behaviour in heavy-duty vehicles

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ABSTRACT

Fuel economy in freight transport is key in attaining economic and environmental efficiency and sustainability. Fuel consumption in Heavy-Duty Vehicle (HDV) is dependent on a set of driver-behaviour parameters and a set of external factors including terrain, loading, vehicle type, and goods etc. Therefore, this study aims at identifying the company-specific critical driving behaviour factors contributing to fuel consumption at HDV fleet using General Estimation Equation analysis. Further, a methodology is proposed to identify fuel-efficient and inefficient drivers using Data Envelopment Analysis and efficiency frontiers. As an illustrative example, a timber haulage company was analysed in which monthly driver behaviour performance data was collected using a telematics system. The results identified average speed, braking, and idling to be key factors influencing fuel consumption. Individual driver's performance in a month is marked as efficient or inefficient considering the set of critical driving-behaviour parameters, and inefficient drivers can improve their performance by following the proposed personalised incentive scheme.

1. Introduction

According to the [European Environmental Agency \(2018\)](#), road vehicles account for 82% of the total greenhouse gas (GHG) emissions from transportation and one-fifth of the European Union's (EU) total GHG emissions. Heavy-Duty Vehicles (HDVs) are responsible for 27% of CO₂ emissions from road transport and almost 5% of the total GHG emissions in EU. In Ireland, the transport sector accounted for the largest share of energy-related CO₂ emissions, with a share of 40% in 2018. Of this transport share, Irish HDV fleet contributes 14% of all road-transport related GHG emissions ([SEAI, 2020](#)). The transport sector is dependent on fossil fuels, and CO₂ emissions from the transport sector are expected to increase in the absence of fuel mix. According to the 'Transport Trends 2019' report by the Department of Transport, Tourism and Sport (DTTAS), expected improvement in the economy would increase activities related to land freight. In the year 2018, a total of ~150 million tonnes of goods were transported by road covering 1595 million kilometres distance ([CSO, 2018](#)). Following the legislative proposal on 17th May 2018 by the [European Commission \(EC\)](#), a new EU regulation 2019/1242 is enforced from 14th August 2019, setting new standards for CO₂ emission for new HDVs. This legislation aims to provide incentives to encourage zero and low emission vehicles. The new regulation also reduces fuel consumption costs for smaller and medium enterprises (SMEs) and consumers and upholds the leadership of EU suppliers and manufacturers in technology.

To reduce emissions from the transport sector, [Huang et al. \(2018\)](#) suggested several measures such as fleet renewal (e.g., EURO VI engines in Europe), improved engine & vehicle technology (e.g., hybrid electric vehicles, particle filters / catalytic / waste heat

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recovery), better fuel quality & use of renewable fuels (e.g., biofuels and higher-octane rating petrol), improved mobility and reduced recurrent traffic congestion, and finally drivers' training. Previous studies indicated relationship between factors such as vehicle speed, acceleration, driving behaviour, traffic condition, road terrain, road grade to fuel consumption and emissions (Liu et al., 2019a, 2019b; Faria et al., 2019; Gallus et al., 2017). In freight transportation, the critical factors are vehicle, environment, traffic, driver and operational characteristics to contribute to fuel consumption (Demir et al., 2014). The driver characteristics which are identified to influence the fuel consumption were driver's aggressiveness, gear selection and idling time. Driving style is considered as the most critical factor in improving fuel efficiency and driver training program in HDV sector could improve fuel economy by 8–10% (McKinnon et al. 2015). Also, Ang-Olson and Schroeer (2002) stated that the driver's training and monitoring were observed to be one of the critical strategies to improve fuel savings. Several review articles stated that driver behaviour could be improved with the help of driver training, education, or eco-driving training, which could improve the fuel economy; however, majority of the articles are focused on passenger cars compared to HDVs (Ang-Olson and Schroeer, 2002; Alam and McNabola, 2018; Demir et al., 2014; McKinnon et al., 2015; Zuraida and Widjaja, 2017; Huang et al., 2018). The driving skills required for HDVs are significantly different from passenger cars because of variation in length, weight, aerodynamics, turning radii, and engine characteristics. As required driving skills differ between passenger cars and HDVs, it is necessary to carry out separate research studies to establish the driving-behaviour factors that impact the fuel efficiency in HDV fleet.

In the road haulage sector, telematics data help to track the vehicles, their fuel consumption, and capture driver's performance measuring harsh acceleration, harsh braking, over speeding, and timely gear change. There are multiple telematics-based fleet management systems in freight sectors such as, Bluetree, Verizon Connect, GPS Insight, Momentum IoT, Webfleet and MyGeotab. Analysis of such data for individual drivers and subsequent advising or additional training could improve driver performance and/or fuel economy. Based on real-world driver-behaviour data, studies have focused on identifying the contributing factors (Walnum and Simonsen, 2015; Ferreira et al., 2015) and haulier's perspective (Liimatainen et al., 2012) to fuel consumption. Driver-behaviour such as, idling (Farzaneh et al. 2020), harsh acceleration-deceleration (Adamidis et al., 2020), fuel consumption using on-board diagnostics (OBD) (Pavlovic et al., 2021) were identified as critical. Recent studies have also worked on developing and verification of fuel consumption models using HDM-4 software (Perrotta et al., 2019), and telematics data (Madhusudhanan, et al., 2020; Bousonville et al., 2020). Walnum and Simonsen (2015) analysed telematics data, driver, vehicle, and infrastructure characteristics of a Norwegian HDV transport company and identified that more than 90% torque power, horsepower and use of the highest gear improves fuel economy. Optimal use of clutch, revolutions (rpm) and idling could reduce the fuel consumption by between 3 and 5 l/100 km (Ferreira et al., 2015).

These driver-behaviour parameters can best be influenced and improved by eco-driving training (Ayyildiz et al., 2017; Díaz-Ramirez et al., 2017; Zavalko, 2018), eco-driving assistance or feedback system (González et al., 2021, Dahlinger, et al, 2018; Strömberg and Karlsson, 2013; Rolim et al., 2014) and eco-routing / best possible route (Zhou et al, 2016; Ahn and Rakha, 2013; Scora et al., 2015; Zeng et al., 2020; Dimokas et al., 2020). One of the downsides of eco-driving is that the benefits reduce over the time; however, recent study by González et al., 2021 observed that the benefits from eco-driving supported by on-board driver assistance technology remain steady over the time.

Also, hauliers transport different types of goods ranging from refrigerated food, raw material, machinery, chemicals, wood, textiles, coal and natural gas, etc. and therefore, each company requires specific sets of axle configuration in trucks ranging from rigid, two to five axles and articulated trucks. It would change the overall fuel economy at the individual level and at a fleet level and hence, identifying the key driving factors contributing to fuel consumption based on the real-world data would help to develop exclusive driver training program at a company level.

Within a haulier company, incentive program could encourage drivers to be fuel-efficient and therefore, helps to improve the fuel economy at fleet level. Incentive programs can be introduced through the implementation of fuel standards and label taxes, registration fees, use of alternative fuel, etc. (Mahlia et al., 2013). McKinnon et al. (2015) suggested increased awareness and incentives to HDV drivers would help to achieve long-term effects of eco-driving. However, an incentive program for professional drivers could be complex exercise because drivers would switch trucks, delivery runs, lack of accurate consumption data, and unfair comparison between drivers (McKinnon et al., 1993; Coyle and Brown, 2004). In terms of methodology, Liimatainen (2011) proposed the eco-driving incentive system for Finnish HDV sector (Buses in Tampere City) by defining fuel economy percentage based on individual driver's performance compared to weighted driver's average consumption in overall 'comparison groups'. These groups were defined based on similar vehicle, routes and time-of-the-day. Based on the hauliers' survey in Finland Liimatainen et al. (2012), concluded that small companies are familiar with energy efficiency actions, but lack knowledge of implementation strategies. Bigger hauliers are more informed and active concerning fuel efficiency than smaller companies.

In incentive programs, driver's efficiency needs to be quantified and compared with other drivers. Data Envelopment Analysis (DEA) is a useful methodology in efficiency evaluation. It has been extensively applied for evaluating efficiency of transport system (Hajduk, 2018). In HDV related literature, DEA was applied in evaluating efficiency of truck performance (Hjalmarsson and Odeck, 1996), trade-off between CO₂ reduction and fatalities in truck transport (Rogers and Weber 2011), the influence of driver-behaviour on safety (Tselentis et al., 2019) and efficiency of truck restriction policies in China (Chen et al. 2019). Further, two-stage DEA was applied in (Wanke and Barros, 2014) to evaluate the efficiency of drivers in the Brazilian motor carrier industry, and Magaña and Organero (2014) proposed a learning system for eco-driving based on imitation of efficient nearby drivers using DEA to assess the driving efficiency for the fuel consumption. The study concluded that this learning by imitation is a more effective method in improving fuel efficiency in comparison with a traditional eco-driving lesson/training.

As discussed in the preceding paragraphs, previous studies have predominately focused on identifying the contributing factors to fuel consumption for passenger car vehicles compared to HDVs. Even fewer studies have proposed incentivising approach for buses or

HDFs. Professional freight drivers have to change their driving behaviour based on the type of goods transported, type of vehicles, and laden weight. External factors such as ambient air temperature, weather conditions, road surfaces (sand, gravel, asphalt, concrete), terrain (flat, hilly, or mountainous), and prevailing traffic conditions are outside the control of a driver but have a direct effect on fuel consumption. From a company's perspective, to improve fuel economy, investigating and improving driver behaviour is a critical consideration as the external factors mentioned are impossible/difficult to control, manage, and improve.

Therefore, the present study develops a framework to determine critical factors influencing fuel consumption in HDFs analysing real driving data and further distinguishes fuel-efficient & inefficient drivers at a company/fleet level for an incentivising scheme. The study utilises average driving behaviour data eliminating the bias due to specific vehicular, environmental, traffic and operational conditions. From a *modelling perspective*, the uniqueness of the present methodology lies in combining a parametric analysis (regression analysis) to identify critical driving behaviour factors and a non-parametric (DEA) analysis to distinguish between inefficient & efficient drivers to develop a framework that can be implemented at a company level to improve fuel economy. Furthermore, the present study is the first ever investigation of freight-telematics data in Ireland., which being a small island country possess distinct transport characteristics.

The rest of the paper includes data collection and processing in Section 2, followed by a brief description of the proposed methodology in Section 3, and the results are presented in Section 4. Further, in Section 5, results are discussed and compared with past research studies and finally, in Section 6 illustrates the conclusion, limitation and future scope for the present research.

2. Methodology

This study proposes a quantitative methodology for detecting driver-behaviour factors contributing to excessive fuel consumption in HDFs, further identify efficient drivers from a group of HDF drivers and eventually develop an incentive scheme to improve driving behaviour to arrive at efficiency. The effect of driver-behaviour factors contributing to fuel consumption was investigated using GEE, which is described in Section 2.4. Efficient drivers were identified using DEA as described in Section 2.3. The proposed incentive scheme is described in Section 2.4 while the overview of the entire framework is described in Section 2.1.

A detailed framework for incentivising efficient driving in HDF drivers in a haulier company was developed. This methodology requires collection and processing of vehicle telematics data which is available for most EURO-VI HDF vehicles and is a common practise in haulier companies. A linear correlation analysis using Pearson Correlation Coefficient (PCC) was carried out to identify the significant driver-behaviour parameters contributing to fuel consumption from the available dataset. If the PCC value between two predictor variables fall outside ± 0.3 at a 95% confidence interval, then only one of those variables was selected as a significant predictor variable for the next stages of the framework.

2.1. Description of the incentivising framework

A *fuel consumption model* which will estimate the fuel consumption of each individual driver based on a set of *key factor variables* was developed using GEE method. Additionally, Driver Efficiency was modelled using DEA method to identify fuel-efficient drivers within

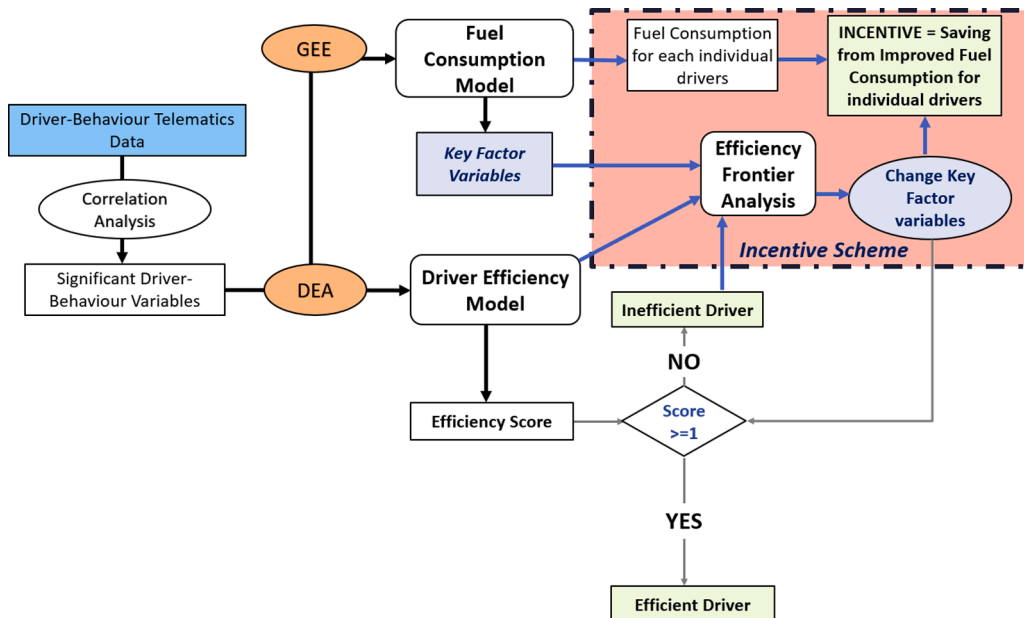


Fig. 1. Flow diagram depicting the incentive scheme.

a haulier company. An efficiency score < 1 indicated inefficient drivers. For these drivers an incentive scheme had been designed with the aim of improving their efficiency. Through an Efficiency Frontier Analysis (EFA) which indicated the change required in *key factor variables* to achieve efficiency, the inefficient drivers could improve their fuel consumption. The monetary savings arising from improved fuel consumption in inefficient drivers undertaking the scheme, would be provided as an incentive to those drivers. This Scheme has been explained in some detail in Fig. 1.

The theoretical details of GEE, DEA model and Efficiency Frontier Analysis are provided in the sections below.

2.2. Generalised Estimation Equation (GEE)

The GEE analysis was used to develop the *Fuel Consumption Model*. GEE method is an extension of the Generalised Linear Model (GLM) to model correlated data and generates unbiased regression coefficients with dependent variable which may not follow a Gaussian distribution (Liang and Zeger, 1986; Ballinger, 2004). GEE is suitable for analysing continuous, ordinal, polychotomous, binary, and count-dependent data. GEE is a marginal method where a monotonic differentiable link function describes the linear relationship between the response variable and its predictors. Link functions such as natural log, reciprocal, and square-root; and distribution functions such as normal, gamma, Poisson, negative binomial are applicable for the dependent variables.

GEE being a marginal mean model and the general form is represented as:

$$g[E(Y_{ij}|x_{ij})] = x'_{ij}\beta \quad (1)$$

where,

x_{ij} is a $p \times 1$ vector of covariates

β consists of the p regression parameters,

$g(\cdot)$ is the link function, and Y_{ij} denotes the j^{th} outcome (for $j = 1, 2, \dots, J$) for the i^{th} subject (for $i = 1, 2, \dots, N$)

Two categories of GEE model were developed, one considering the significant variables identified using correlation analysis (*Key Factor Model*) and the other considering all measured predictor variables using backward elimination technique (*Backward Elimination Model*). While the later considered each predictor variable for its marginal impact, the former (*Key Factor Model*) quantify the influence of significant predictor variables on the response variable (fuel consumption) while minimising the effect of multicollinearity (Gujarati et al., 2012). All variables which were significant at 95% confidence interval ($p\text{-value} > 0.05$) were included in the final formulation of both the GEE models. Two goodness-of-fit criteria, Quasi Likelihood under Independence Model Criterion (QIC) and Corrected Quasi Likelihood under Independence Model Criterion (QICC) parameters Pan (2001) which are similar to Akaike Information Criterion (AIC) but do not include consideration of likelihood function were utilised to choose the best-fit model within a certain category.

The GEE models were developed using 80% of the available data for training purposes, while the remaining 20% were used for validation of the models. The performance of the GEE models was validated using Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Percentage Error (MPE) and the model with best predictive performance was chosen for modelling Fuel Consumption.

2.3. Data Envelopment analysis (DEA)

For a haulier company, fuel consumption is the most important factor for profit margins. The driving-behaviour of professional HDV drivers influencing fuel consumption and driver efficiency can be modelled based on HDV drivers' behaviour. The DEA method was used to estimate the relative efficiency of every driver within a group. Driver efficiency is defined as the driver's ability to cover as much distance as possible, travelling at appropriate and permitted average speed, with minimum fuel consumption, while optimising other driver-behaviour factors affecting fuel economy (such as, overrevving, idling, harsh braking, hard acceleration and deceleration).

DEA is a non-parametric method introduced by Charnes, Cooper and Rhodes (CCR) (Charnes et al., 1978). Since the introduction of the first model CCR, it has been widely used to evaluate the relative efficiency of decision-making units (DMUs). The CCR, BCC (Banker, Charnes and Cooper) (Banker et al., 1984), and additive (ADD) (Charnes et al., 1985), radial DEA model, slack-based measure (SMB) model (Tone, 2001), and non-radial DEA (NR-DEA) model are the most known traditional DEA models. NR-DEA models are useful in avoiding all disadvantages of traditional and radial DEA models related to remaining slacks, overestimation score of efficiency, discriminatory power, and absence of preference weights and has been adopted in this study (Rashidi et al., 2015). These models group DMUs into inefficient and efficient categories using relative efficiency score where an efficient driver has a score ≥ 1 . Most traditional DEA models do not distinguish inputs and outputs as desirable and undesirable. But, in real production processes undesirable inputs can produce undesirable outputs. So, both desirable and undesirable inputs and outputs are presented in the DEA methodology described here. The joint production process where desirable inputs and undesirable inputs are denoted as x and e and desirable outputs and undesirable outputs are denoted as y and u , can be presented as:

$$T = \{(x, e, y, u) : (x, e) \text{ can produce } (y, u)\} \quad (2)$$

The joint production theory by Wu et al. (2017) provides the following optimisation formulation for the NR-DEA,

$$\min W_l \frac{1}{L} \sum_{l=1}^L \theta_l + W_j \frac{1}{J} \sum_{j=1}^J \theta_j \quad (3)$$

s.t.

$$\sum_{k=1}^K \lambda_k x_{nk} \leq x_{n1}$$

$$\sum_{k=1}^K \lambda_k e_{lk} \leq \theta_l e_{l1}$$

$$\sum_{k=1}^K \lambda_k y_{mk} \geq y_{m1}$$

$$\sum_{k=1}^K \lambda_k u_{jk} = \theta_j u_{j1}$$

$$\lambda_k \geq 0$$

$$\theta_k^l \leq 1$$

$$\sum W_l + W_j = 1$$

$$n = 1, \dots, N.$$

$$l = 1, \dots, L.$$

$$m = 1, \dots, M.$$

$$j = 1, \dots, J.$$

$$k = 1, \dots, K.$$

where, K number of DMUs produced M desirable outputs $y_m (m = 1, \dots, M)$ and J undesirable outputs $u_j (j = 1, \dots, J)$ using N desirable inputs $x_n (n = 1, \dots, N)$ and L undesirable inputs $e_l (l = 1, \dots, L)$. The unified efficiency of this model varies between 0 and 1. The unified efficiency of two efficiency scores θ_l and θ_j is measured using different non-proportional adjustments and decision-maker specified weights W_l and W_j .

Driver efficiency evaluation has been conducted using two NR-DEA models where driver performance per month was used as DMU. Similar to the *Fuel Consumption GEE Model*, the first model considers the variables identified using correlation analysis, while the second model considers all variables. All applied inputs and outputs were categorised as desirable and undesirable variables for both models, mainly based on the correlation factor. Undesirable variables have been grouped as inputs and outputs that negatively impact fuel consumption, while factors that have to be maximised as a result of fuel efficiency were categorised as desirable variables. The

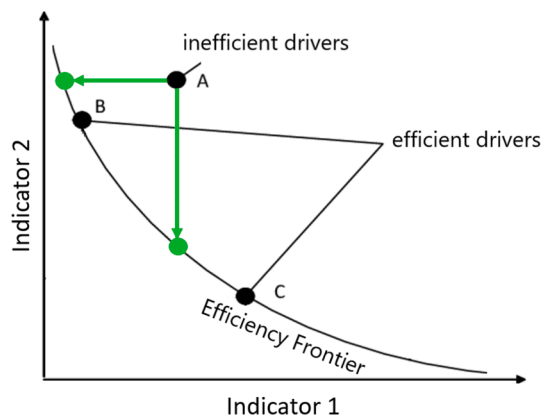


Fig. 2. Efficiency frontier with two indicator variables.

model which was able to better discriminate between efficient and inefficient drivers was selected for further Efficiency Frontier Analysis.

2.4. Efficiency frontier analysis

Efficiency Frontier Analysis (EFA), as a concept of DEA, is a useful tool for interpreting and comparing the efficiency of different driver efficiency. Since the publication of papers [Charnes et al. \(1978\)](#), [Banker et al. \(1984\)](#) and [Fare et al. \(1983\)](#), the literature related to the frontier analysis in DEA began to develop. Most recently, [Daraio et al. \(2020\)](#) provided systematic review of the application of frontier methodology in different sectors, including transportation. EFA shows the efficiency status of DMUs with respect to a smaller set of variables and indicates the quantitative change that needs be made to achieve efficiency. All efficient DMUs can be presented using efficiency frontiers where DMUs with efficiency score equal to 1 form a curve ([Fig. 2](#)) ([Ramanathan, 2003](#)). Inefficient DMUs are not on the frontier curve but can reach the frontier by satisfying the objective function and constraints. Consequently, the efficiency frontier is a suitable tool for rationally dividing drivers on two categories i.e., efficient and inefficient, according to their distance to the frontier. Further, the frontier shows the curve of maximum output with different input combinations or the minimal combination of necessary inputs for different outputs. In [Fig. 2](#), a convex efficiency frontier of two inputs and one output is shown. The ratios of input to output are used as axes and denoted as Indicator 1 and Indicator 2. [Fig. 2](#) shows that a driver who consumes less input per unit of output lies on the curve. Consequently, drivers B and C are denoted as efficient drivers, while driver A is not efficient.

The efficiency frontier or production frontier is identified for a production possibility set. The production function defined in Eq. (3) defines the outputs in terms of all the inputs and identifies the production frontier for the production possibility set described in Eq. (2) ([Amirteimoori et al., 2006](#)). Frontiers for the NR-DEA model should be presented as ratios of outputs to inputs. In traditional CCR and BCC type DEA models, frontiers are created with drivers with best output to input ratio. This frontier is known as envelope and is generally developed in the case of one output and input ([Benicio and Soares de Mello, 2015](#)). However, with desirable and undesirable factors in DEA model, the efficiency frontier has to include them as ratios of *undesirable output/undesirable or desirable input* and *desirable output/undesirable or desirable input* (efficiency frontiers in [Fig. 7](#)). An undesirable output such as fuel consumption is minimized in production function and production/efficiency frontiers created from ratios of undesirable output to inputs are right-angled and concave towards origin. The indicators identified as most influential for fuel consumption were further investigated using efficiency frontiers. The inefficient drivers who were not on the efficiency frontier in a particular month could improve the status

Table 1
Specification of heavy-duty vehicles at fleet level.

Model (# of trucks)	Model year & Euro Class	Engine & Power	Engine displacement (Litres)	Drag coef.	Frontal area (m ²)	Torque	After-treatment system	Transmission	Weight (kg)
Scania G 450 (5)	2016, Euro 6	6-cylinder engine; 450hp @1900 rpm	12.7	0.62	8.14	2350Nm @1000–1300 rpm	Selective Catalytic Reduction (SCR) with catalytic converter	Scania GRSO905R Opticruise Overdrive 14 speed with fully automatic clutch	8985
Scania G 440 (3)	2013, Euro 5	6-cylinder engine; 440hp @1900 rpm	12.7	0.62	8.14	2300Nm @1000–1300 rpm	SCR with catalytic converter	Scania GRSO905R Opticruise Overdrive 14 speed with fully automatic clutch.	8990
Scania R 450 (3)	2014, Euro 6	6-cylinder turbo diesel engine; 450 hp @ 1900 rpm	13 l	0.75	8.31	2350Nm @1000–1300 rpm	SCR with catalytic converter	Scania GRS895 Opticruise automatic with retarder	9660
Scania R 500 (2)	2019, Euro 6	6-cylinder turbo-diesel engine; 500hp @ 1900 rpm	13 l	0.75	8.31	2550Nm @ 1000–1300 rpm	SCR with catalytic converter	Scania GRSO905R Opticruise 14-speed auto with the Scania R3500 five-stage retarder	9730
Renault T 480 (5)	2015, Euro 6	6-cylinder engine; 470 hp (353 kW) from 1404 to 1800 rpm	12.8 l	0.71	8	2400Nm @ 950 rpm to 1404 rpm	Compact after-treatment system (catalytic converter, Diesel Particulate Filters (DPF), SCR)	Optidriver AT 2612F gearbox (14.94–1) with automatic clutch.	7500
Volvo FH 460 (4)	2010, Euro 5	6-cylinder engine; 460 hp @ 1400–1900 rpm	12.8 l	0.59	8.1	2300Nm @ 1000–1300 rpm	Combines Diesel Oxidation Catalysts (DOC), DPF, SCR and Ammonia Slip Catalyst (ASC).	14-speed splitter and range manual gearbox.	9250

by moving on the frontier curve by changing indicator variables following the directions of the maroon arrows shown in Fig. 2.

3. Driver behaviour data

A timber haulage company located in the north-western part of Ireland was considered for this study. The company collects wood from forest areas located in north-western parts of Ireland and distributes to sawmill companies located all over Ireland and transports around 3000 tonnes of timber every week. The company has 22 trucks, 15 of them are Euro-VI and 7 of them are Euro-V diesel engine trucks along with 30 trailers and employs 30 male professional drivers. Table 1 presents the specification of all 22 vehicles describes model year, euro class, engine displacement, drag coefficient, torque, after-treatment system, transmission, and weight. Further, the company has also installed Bluetree (BT 500) RCOM Fleet Management System installed in all their trucks & trailers. The company claims that the telematics system reduced fuel consumption and operating costs by 20%. All trucks and trailers have Continental ContiPressure Check System; a direct measurement system mounted inside the tyre to monitor air pressure and tyre temperature. If the inflation pressure drops, the driver is informed, which would reduce tyre damage and blowouts. Overall, based on engine and power standards, frontal area, torque, all 22 trucks have similar vehicle specifications.

In terms of driving patterns, drivers in the selected haulage company start their journey on Monday morning from Leitrim and travel to client-specific locations across Ireland. Drivers typically take motorways (speed limit 120 Km/h) and national primary roads (speed limit 100 km/h). The delivery locations are typically located within 15 km of motorways and national primary roads. Therefore, drivers have to drive around 30 km on secondary or rural roads, increasing their fuel consumption. Lastly, Ireland has flat terrain conditions, low-lying area in the midlands, and ringed by mountain ranges. The sinuosity index for motorways and national primary roads indicates that most of these roads are straight roads.

As driver-behaviour changes with the type of goods, laden weight, and each company hauls a specific type of goods, laden weight and, therefore, driving-behaviour and required driving skills changes with companies. Nowadays, companies are adopting telematics system to track their vehicles and fuel consumption. In this system, Telematics Control Unit (TCU) collects vehicle information via CAN-BUS port (controller area network), GPS and transfers the collected information to the Cloud Server via LTE, cellular or GPRS. The data stored in Cloud Server is processed and available to company owners/managers in the form of web or mobile applications. Driver-behaviour parameters such as fuel consumption, distance, average speed, idling, hard acceleration, hard deceleration, over-revving, coasting, the number of stops, and brakes are provided to the end-user in the form of web/mobile application. Definitions for some of the key variables collected by telematics system are as follow.

- Average speed is defined as the total distance travelled by each trip divided by the total time taken to complete the journey.
- Hard acceleration and hard braking: These features are time localized, and it is an event when more force than usual is applied to the vehicle's brake or accelerator. Typically, each telematics system sets minimum and maximum threshold values to define hard acceleration and hard braking. For example, typically, in heavy-duty vehicles, the minimum and maximum thresholds for hard acceleration are 7.76 km/h/s (0.22 g) and 28.24 km/h/s. Minimum and maximum thresholds for hard braking are defined as 7.76 km/h (0.22 g) and 35.29 km/h. In this present study, the variables are defined as the percentage of time travelled in harsh acceleration and the number of harsh brakes respectively.
- Coasting is the process in which a vehicle moves with either the clutch depressed or gear in neutral or both at the same time. Vehicle speed and the accelerator position provide information about coasting. It is defined as the percentage of the total distance travelled in coasting.
- When the current engine RPM exceeds the present engine speed threshold, it is defined as overrevving. Further, when the engine is on, and the vehicle is not moving for a predefined period of time, it is defined as idling. Both are defined as the percentage per time.

In practice, the incentive programs are executed by comparing monthly performance reports rather than by comparing every individual trip. Therefore, monthly driver-behaviour telematics data collected over a 10-month period from January to October 2019 were considered for this study. Table 2 represents the descriptive statistics of the studied dataset.

Table 2
Summary of telematics data collected from the timber company.

Variables (Unit)	Min.	Max.	Mean	Std. Deviation
Avg. fuel Consumption (litres/100 km) per month	33	98.8	49.6	8.9
Driver's Age (years)	25	55	42.8	9.9
Total Distance per month (km)	33.4	12820.2	7621.3	3772.4
Total Duration per month (hrs)	1.32	256.35	156.35	68.56
Avg. Speed (km/h)	7.0	64.0	45.9	13.2
Over-revving (%)	0.1	51.4	18.5	16.4
Harsh Brake (%)	0.0	45.0	7.2	9.5
Idling (%)	0.5	26.9	6.4	4.1
Top Gear (%)	0.0	58.9	15.0	19.8
Hard Acceleration (%)	39.3	79.0	62.8	6.7
Hard Deceleration (%)	6.2	29.6	19.2	4.2
Coasting (%)	0.0	75.7	9.3	10.0
Brake/100 km	80.0	501.9	157.0	65.7

Table 3

Pearson correlation analysis.

Parameters	Driver's Age	Avg. Speed	Distance	Overrevving	# of Harsh Brakes	Idling	Top gear	Hard Accl.	Hard Decl.	Coasting	#of brakes	# of Stops
Fuel Consumption	−0.06	−0.78**	−0.54**	−0.24**	−0.12	0.12	−0.2*	0.34**	0.12	−0.36**	0.78**	−0.32**
Driver's Age	1	0.03	0.02	0.1	0.08	0.03	−0.09	−0.09	−0.12	0.15	0.06	0.02
Avg. Speed		1	0.76**	0.42**	0.25**	−0.04	0.16	−0.028**	0.12	−0.01	−0.72**	0.42**
Distance			1	0.29**	0.34**	−0.29**	0.24**	−0.3**	0.08	−0.15	−0.62**	0.82**
Overrevving				1	0.24**	0.08	−0.18*	−0.12	−0.04	−0.14	−0.24**	0.16
# of Harsh Brakes					1	−0.17*	−0.23**	0.23**	0.34**	−0.11	−0.09	0.32**
Idling						1	−0.24**	0.12	0.16	−0.17*	0.18*	−0.35**
Top gear							1	−0.53**	−0.39**	0.19*	−0.21*	0.25**
Hard Accel								1	0.77**	−0.16*	0.41**	−0.29**
Hard Decel									1	−0.47**	0.2*	−0.01
Coasting										1	−0.28**	−0.18*
# of brakes											1	−0.46**

Note: '***' indicates correlation is significant at the 0.01 level and '**' indicates correlation is significant at the 0.05 level; '−' sign indicates negative correlation and '+' sign indicates positive correlation.

Fuel consumption at fleet level varied between 33 l/100 km to 98.8 l/100 km. Some of the HDVs collect timber from deep inside the forest areas. Due to the rough terrain conditions, the maximum fuel consumption observed was very high at around 98 l/100 km and the minimum average speed observed was around 7 km/h. Average age of the drivers is 42.8 years and ranging between 25 and 55 years. Average distance travelled by drivers in a month is around 7621 km and mean driving duration in a month is around 156 h.

A linear analysis between fuel consumption and average speed was observed to be exponentially decreasing and HDVs were observed to be fuel efficient at 60kmph. As per the Road Safety Authority (RSA) in Ireland, driving hours are regulated to avoid fatigue-related collisions. EU law states that drivers of good vehicles >3.5 t (including trailers) should not drive more than nine hours per day or 56 h per week or 90 h in two consecutive weeks. It also recommends drivers to take a mandatory break of at least 45 min after every 4.5 h. However, there are no recommended guidelines about driving distance per day or week and consequently, a significant variation is observed for this variable. Driver-behaviour such as overrevving, idling, hard acceleration, hard deceleration, and coasting are represented in percentages representing time durations of these activities in a trip. Typically, drivers tend to coast while driving downhill and during coasting, there is less control as the engine is not connected with wheels and could increase the risk of collision substantially. Idling percentage indicates the time the engine is running in idle position as a fraction of the total operating time. Harsh and regular brakes were both measured as numbers per 100 km. High values of hard acceleration were observed as the trucks sometimes travelled on rough terrain for this haulage company.

The study focused on identifying critical driving behaviour factors contributing to fuel consumption using GEE model and identifying fuel-efficient drivers using DEA model. As mentioned earlier, all 30 male professional drivers' monthly performance reports for ten months of data were considered for this study. For identifying critical factors, we analysed the data of all 30 drivers. However, not all 30 drivers have worked for all ten months, which is typical in any haulage company. Some drivers worked for only one month and did not work for other months. Therefore, total data sample size is 173, which was used to identify critical driving behaviour factors contributing to fuel consumption using GEE. The GEE models were developed using 80% of the available data for training purposes, while the remaining 20% were used for validation of the models. In identifying fuel-efficient drivers, a total of 9 drivers who worked consistently for the entire ten months period under consideration (sample size 90) were selected for the analysis.

4. Analysis and results

In this section, development of the *Fuel Consumption Model* and the *Driver Efficiency Model* were described and the results of the utilising the entire framework described in Fig. 1 on the drivers of the haulage company are discussed.

4.1. Correlation analysis

A linear relationship between parameters was investigated by Pearson correlation and presented in Table 3. In the table, positively correlated variables were marked with green while negative correlations were marked red, while the colour intensity changed based on the cell values. Fuel consumption was chosen as the response variable for both GEE models and the undesirable output for both DEA models.

4.2. Fuel consumption modelling using GEE

The effect of the driver-behaviour variables, shown in columns 2–12 in Table 3, on the monthly fuel consumption (F_C) in HDV fleet of the haulage company was modelled using GEE. The 'Backward Elimination Model' considered all 11 measured predictor driver-behaviour variables in which insignificant predictor variables (p -value > 0.05) with the highest p -value were removed one by one.

The second model, the 'Key Factor Model' considered a set of predictor variables that were independent (not correlated) of each other. Predictor variables were said to be correlated if computed PCC is less than -0.3 or greater than 0.3 . Therefore, if two predictor variables are correlated, then only one predictor variable was considered in the Key Factor Model. High correlation was observed

Table 4
GEE Models using Telematics Data.

Parameters	Backward Elimination Model			Key Factor Model		
	Coef.	Std. Error	p -value	Coef.	Std. Error	p -value
Avg. Speed	−0.008	0.001	< 0.01	−0.010	0.001	< 0.01
Overrevving (%)						
# of Harsh Brakes				0.002	0.001	0.035
Idling (%)				0.004	0.002	0.016
Coasting (%)	−0.006	<0.001	< 0.01			
Brake (per 100)	3.77E-04	<0.001	< 0.01			
Stops						
Intercept	4.267	0.043	< 0.01	4.314	0.076	< 0.01
QICC (model fit)	8.693			9.235		
MAPE (validation)	17%			15%		
RMSE (validation)	10.54			8.70		
MPE (validation)	−11%			−9%		

between average speed & distance, number of brakes & stops and hence, only one variable (i.e. average speed) was selected in the 'Key Factor Model' to minimise the multicollinearity. Likewise, a high correlation existed between hard acceleration and hard deceleration (0.773) and hence, only one variable was selected in the key factor model. A similar situation existed between parameters brakes (per 100 km) and the number of stops. Eventually, age, average speed, number of harsh brakes, idling, coasting and hard acceleration was considered in the second model.

GEE models were developed using SPSS software. Log-link with gamma distribution model was found to be the best-fitted model for the data used in this research (Table 4). The equation of the two best-fit models are as follows:

$$\ln(F_C) = 4.267 - 0.008 * \text{Avg.Speed} - 0.006 * \text{Coasting} + 3.77E - 04 * \# \text{ of brakes} \quad (4)$$

$$\ln(F_C) = 4.314 - 0.010 * \text{Avg.Speed} + 0.002 * \# \text{ of Harsh Brakes} + 0.004 * \text{Idling} \quad (5)$$

Lower MAPE, RMSE, and MPE and closer to zero indicate better accuracy of model. Considering the predictive performance (MAPE, RMSE, and MPE) on the validation dataset as shown in Table 4, the 'Key Factor Model', described in Eq. (5) was chosen as the appropriate *Fuel Consumption Model* developed using GEE. The validation performance of the model was shown in Fig. 3, in which the coefficient between predicted and observed fuel consumption was observed to be 1.03 and follow a linear trendline with zero intercept. It indicates that the *key factor model* can predict the observed fuel consumption with only 3% of error at $R^2 = 0.97$.

The results indicate that average speed negatively impacted F.C. Fuel consumption exponentially decreases by 10% for every 10 kmph increase in the average speed of HDVs and exponentially increases by 2%, and 4% for ten number of harsh brakes and 10% increase in idling, respectively.

4.3. Driver efficiency modelling using DEA

Similar to GEE analysis, two DEA formulations were developed to find the efficiency of drivers. The first model, '*All Factor Model*' considered all the variables with correlation value ($PCC \geq \pm 0.3$) in relation to fuel consumption as shown in Table 3. Variables, such as harsh brake and top gear were not considered as indicators for DEA models because they included zero values. Driver performance for 9 drivers per month for 10 months -i.e., 90 samples were considered as DMUs.

Average speed, overrevving, idling, hard acceleration, hard deceleration, coasting, number of brakes and stops were considered as inputs. All variables with negative environmental impacts, such as fuel consumption, and variables which can increase fuel consumption such as overrevving, brake and idling were grouped as undesirable variables, while variables otherwise are grouped as desirable variables. The developed DEA model aims at minimising undesirable inputs and output, which are overrevving, idling, hard acceleration and deceleration, brakes and fuel consumption, respectively.

The second DEA model is the '*Key Factor Model*' where PCC were used in selecting the most significant desirable and undesirable inputs and outputs as shown in Tables 3 and 5. The results are shown in Figs. 4–7. Individual driver's overall driving performance every month is presented as a DMU. The drivers studied were denoted as A to I. Fig. 4 shows the results of '*All Factor Model*' and driver "B" was the most efficient driver for most of the months considered, while Driver "A" was the least efficient. Driver "E" and "F" were efficient in 4 months followed by Driver "I" in 3 months, drivers "C", "D" and "H" were efficient in 2 months and driver "F" was efficient only for one month. The area of the efficiency polygon gave good estimation of consistency of efficiency. In Fig. 4a the monthly distribution of efficiency of drivers in '*All Factor Model*' can be seen. For example, from Fig. 4a can be seen that driver "A" had efficiency of around 80% in two months, while driver "B" has been 100% efficient in five months.

The results of the *Key Factor Model* (Fig. 5) differ from *All Factor Model*. In Fig. 5a, the monthly distributions of drivers' efficiency for *Key Factor Model* have been plotted. The most efficient for the most months is the Driver "I" followed by Drivers "C" and "G" who are efficient in three months, while drivers, "B" and "D" are efficient in two months followed by Driver "E". Drivers with the least efficiency scores are "F" and "H". Efficiency scores of *Key Factor Model* are lower in comparison to *All Factor Model* showing better variability using fewer indicators.

The initial analysis using the *Key Factor DEA Model* distinguished the efficient drivers from the inefficient ones. Further investigation regarding the possibilities of efficiency improvement in inefficient drivers was carried out using sensitivity analysis of the

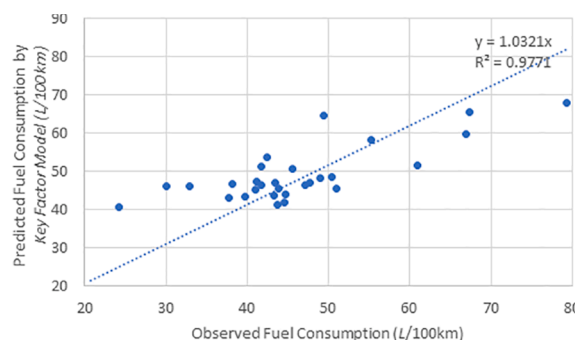


Fig. 3. Validation performance of the key factor model.

Table 5
Selected variables for DEA models.

Indicators	All Factor Model	Key Factor Model	Classification of variables
Fuel Cons. (l/100 km)	✓	✓	UDO*
Total Distance per month (km)	✓	✓	DO*
Avg. Speed (kmph)	✓	✓	DI*
Over-revving (%)	✓		UDI*
Harsh Brake (%)			
Idling (%)	✓	✓	UDI
Top Gear (%)			
Hard Acceleration (%)	✓	✓	UDI
Hard Deceleration (%)	✓		UDI
Coasting (%)	✓	✓	DI*
Brake/100 km	✓	✓	UDI
Driver's Age (years)	✓	✓	DI
Stops	✓	✓	DI

*DI-Desirable Input; *UDI-Undesirable Input; *DO-Desirable Output; *UDO-Undesirable Output.

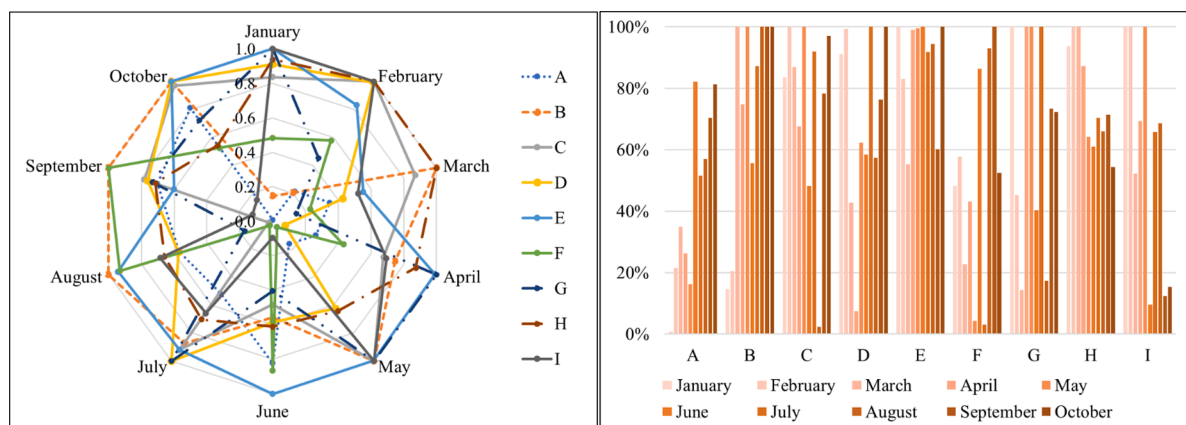


Fig. 4. Efficiency of drivers in all factor DEA model.

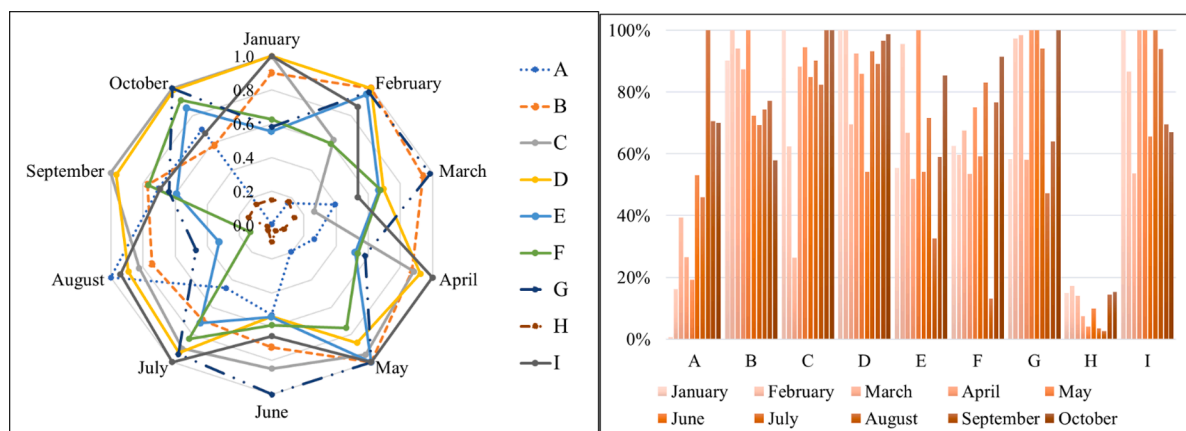


Fig. 5. Efficiency of drivers in key factor DEA Model.

model. In this step, all efficient drivers were eliminated from further analysis, and investigation was carried out only for inefficient drivers. In the first step, the average speed was increased by 10% in Key Factor Model, while other factors were unchanged. The fuel efficiency was improved for all drivers (see Fig. 6). In a subsequent step, the brake was reduced by 20% with original speed and the efficiency of many inefficient drivers were improved even further. This analysis showed the need of developing customised driver efficiency management plan for individual drivers based on their unique driving behaviour.

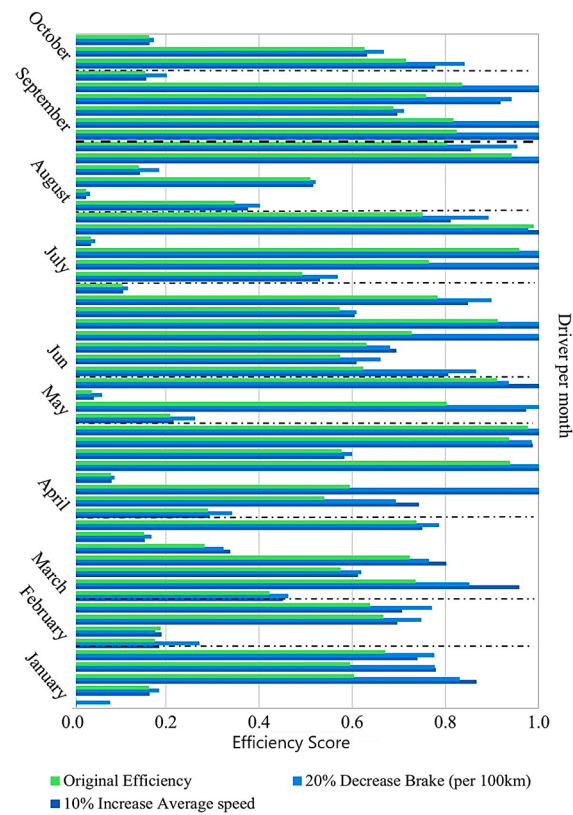


Fig. 6. Effect of variations of average speed and brake on the fuel efficiency.

4.4. Efficiency frontier analysis for incentivising drivers

EFA is another DEA concept used for identifying the fuel-efficient drivers with a smaller set of indicators. Based on EFA, other drivers' potential to become efficient can be recognised by distance from the fuel-efficient drivers. Here, the GEE and Pearson correlation results were used as the inputs for the DEA model, which is the next step of the incentive framework. Using two DEA models i. e., *All Factor Model* and *Key Factor Model* fuel-efficiency of drivers have been obtained. The EFA was carried out considering 'Key Factor Model' developed by GEE and DEA methods. Fig. 7 shows that only a small number of drivers are efficient i.e., form the frontier curve, while several of them are far from the frontier. This is because only some drivers satisfy ratios of UDO with DI or UDI and DO with DI or UDI. For example, in the first plot, only one driver is on the frontier because the driver has the minimum value for ratios of fuel consumption/# of stops and fuel consumption/# of brakes, which should be minimized to be efficient. Otherwise, for ratios distance/# of stops and distance/# of brakes, which have to be as high as possible, three drivers can be found on the frontier. As described in Section 2.4, frontiers are created as a ratio of undesirable output with two inputs and desirable output with two inputs. Since drivers should reduce undesirable output to become efficient, they have to move towards the origin. Therefore, the frontier has a form of a convex function. The frontier's concave shape is related to desirable outputs because drivers have to increase it as much as possible to improve efficiency in efficiency evaluation.

The GEE model identified three critical determinants of fuel consumption: average speed, idling and number of harsh brakes. The effect of these factor on driver performance and possible efficiency set were explored using EFA in Fig. 7. Efficiency frontiers of inputs with respect to fuel consumption (UDO) and the efficiency frontiers with respect to distance (DO) are shown. From Fig. 7, it can be noted that efficiency frontiers with respect to distance included two or three drivers while the performances of other drivers are lower and far from frontiers. Efficiency frontiers with respect to fuel consumption included more drivers and other drivers are closer to the frontier. This means that more drivers are efficient or nearly efficient. For the inefficient drivers to become efficient, they need to change one or two of the studied critical factors in the direction of the frontier. This is the basis of the proposed incentive scheme and is described with a case study in the following section.

4.5. Case Study: Example of incentive scheme identification

The incentive scheme is based on the concept that the performance of the inefficient drivers who are away from the efficiency frontier, can be improved by changing critical factors to the extent as indicated from the EFA. In Fig. 8 an extract of Fig. 7 is shown with two 'key factors' average speed and idling where the first driver (X), denoted with a red circle, is away from the efficiency frontier with

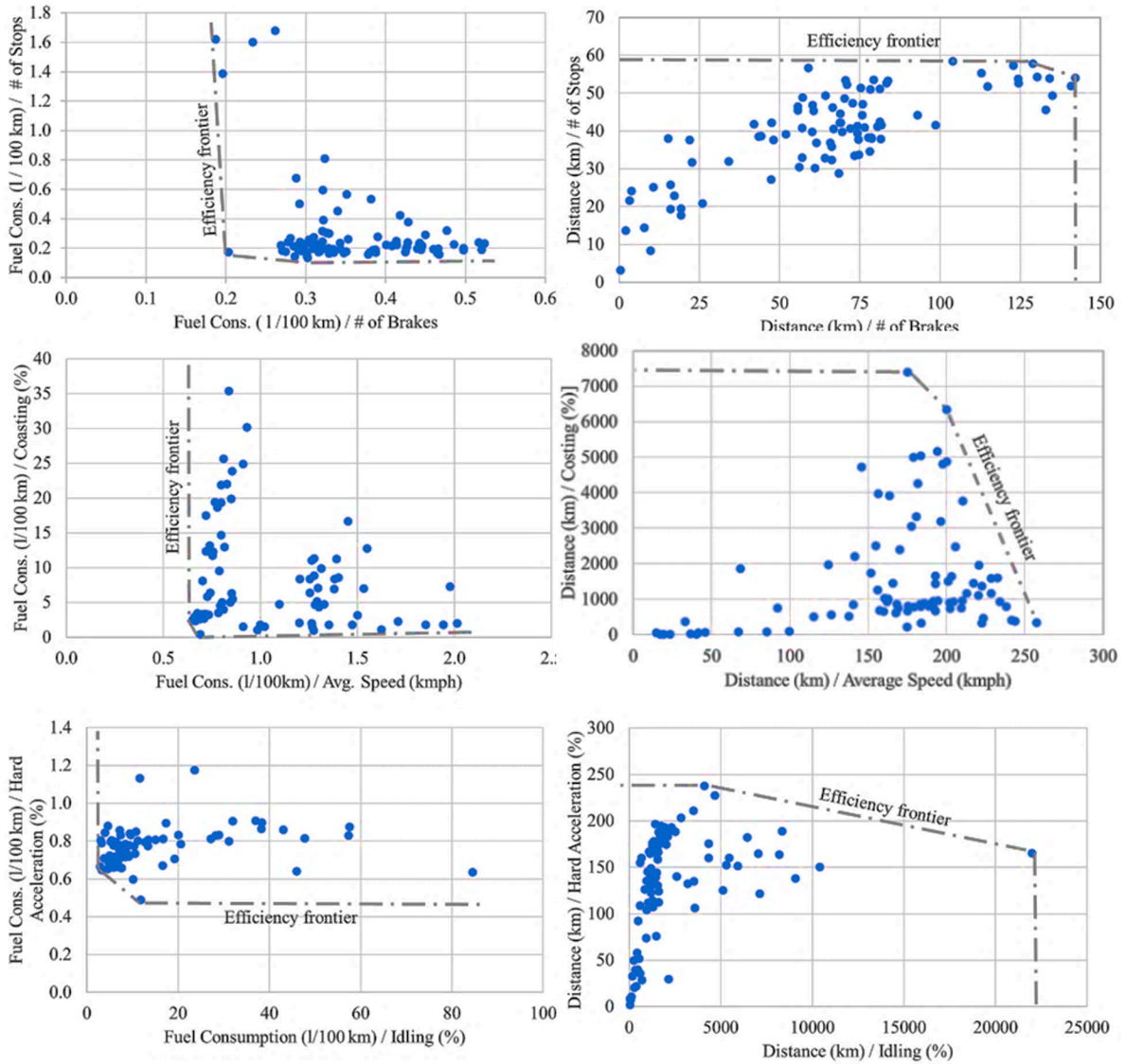


Fig. 7. Efficiency frontier with respect to fuel consumption and distance.

a lower average speed and higher idling in comparison with other drivers on the frontier. For driver X, using GEE Eq. (5), fuel consumption is calculated as 58.79 l/100 km.

$$F_C = \exp[4.314 - 0.010 \times (28) + 0.002 \times (8) + 0.004 \times (6)] = 58.79 \text{ l/100 km}$$

To improve driver's fuel efficiency, driver X needs to change the idling from 6% to around 2% (shown using green arrow), which would reduce fuel consumption to 57.85 l/100 km by keeping all other variables unchanged. Likewise, fuel consumption of driver 'X' can be reduced to 53.73 l/100 km by increasing the speed to 35kmph and reducing idling to 1%, i.e.,

$$F_C = \exp[4.314 - 0.010 \times (28) + 0.002 \times (8) + 0.004 \times (2)] = 58.79 \text{ l/100 km}$$

$$F_C = \exp[4.314 - (0.01 \times (35) + 0.002 \times (8) + 0.004 \times (1))] = 53.73 \text{ l/100 km}$$

Nevertheless, the inefficient drivers can become fuel efficient either by increasing the speed or by reducing the idling or both. The savings from unused fuel can be paid as incentive to the driver X.

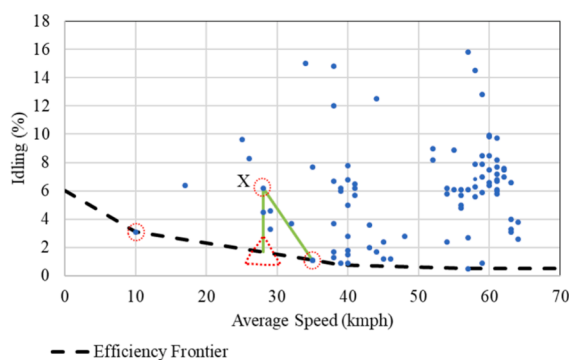


Fig. 8. Efficiency frontier based incentive process.

5. Discussion

This study proposes development of a case-specific incentive framework for improving fuel efficiency in HDV drivers in a haulier company utilising real telematics data.

For both GEE and DEA analysis, 'Key Factor Model' involving fewer number of parameters than the other model (involving all factors) was found to be the most effective. GEE analysis indicated that average speed, idling and number of harsh brakes were increasing fuel consumption. These contributing factors were consistent with previous research studies (Walnum and Simonsen, 2015; Ferreira et al., 2015). As per the results, drivers in this timber-haulage company can be advised to drive at 60kmph for best fuel efficiency. The DEA analysis, 'Key Factor Model' involved additional factors which were identified using PCC analysis including hard accelerations, coasting, drivers age and stops. The model identified the comparative efficiency of the drivers studied. Some of these drivers were significantly more efficient than others with low fuel consumption and high distance coverage. The variability of driver efficiency can be seen in Figs. 4 and 5. For the inefficient drivers increase in average speed or decrease in braking improve efficiency in most cases as shown in Fig. 6. However, such overall measures cannot ensure to equal improvement of efficiency of all drivers. EFA analysis showed that comparatively small reduction in number of stops and idling can improve the possibilities of being on the efficiency frontier in relation to fuel consumption as an undesirable output (Fig. 7). In relation to distance, the relationships are more complex, and it seems small increase in average driving speed can improve driver efficiency.

Eventually a targeted incentive method is proposed combining the GEE and DEA through an efficiency frontier-based analysis. The incentive scheme is customised for each individual driver and is determined by measured change in a set of driver-behaviour parameters. While implementing the incentive scheme, the inefficient drivers who are improving their driving performance should be rewarded with incentives to encourage them and efficient drivers should be rewarded with fixed incentives to motivate them continuing to drive efficiently. This scheme is customised for a haulage company and personalised for individual drivers which is expected to improve chances of success.

6. Conclusion

This paper provides a quantitative tool to identify driver behaviour parameters influencing eco-driving efficiency by analysing real-driving telematics data and proposes a framework for an incentive scheme to promote eco-driving amongst professional HDV drivers. The study is one of the first in Ireland. Ten months of telematics data were collected from a timber haulier company located in the Western part of the country. Pearson correlation analysis, GEE and DEA methods were investigated, and two GEE and two DEA models were developed to fulfil the objectives.

The proposed methodology could be used to identify the company, type of work and driver-specific, driver-behaviour factors to improve the fuel economy at the fleet level. Also, it would help individual drivers to improve their fuel economy by identifying the important driving behaviour parameters they need to improve to become fuel-efficient. At company-specific level, professional HDV drivers could be advised or trained to avoid harsh brakes, idling, continuous use of brakes, and hard acceleration in order to improve fuel economy. Further, drivers could be advised to drive at a speed of around 60kmph to improve the fuel economy. Development of a specific GEE model for an individual haulier for specific goods established the need for considering a set of factors or driver-behaviour variables in modelling fuel consumption rather than focussing on fuel consumption as the only factor in developing an incentive scheme.

The DEA analysis identified that the drivers show widely different efficiency levels while working for the same company and delivering same goods but at different destinations, driving different distances and on different routes. A company-wide policy such as 10% increase in speed or 20% reduction in brakes may improve driver efficiency but cannot be guaranteed to have similar effects on all drivers. A driver-specific scheme is more appropriate and has been proposed in this study. The proposed methodology can be applied to other haulage company to understand the specific driver training factors and incentivising program. Investigating and comparing the results with other haulage company and considering external factors such as ambient air temperature, weather conditions, road surfaces (sand, gravel, asphalt, concrete), terrain (flat, hilly or mountainous), traffic condition on roads merit a deeper investigation.

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